

The Solar System

Study Guide & Questions

Overview

The Solar System consists of the Sun and everything that orbits around it, including eight planets, their moons, dwarf planets, asteroids, comets, and other celestial objects. The Sun contains 99.86% of the Solar System's mass and provides the gravitational force that holds everything together.

The Eight Planets

Inner Planets (Terrestrial): Mercury, Venus, Earth, Mars - These are rocky planets with solid surfaces.

Outer Planets (Gas Giants): Jupiter, Saturn, Uranus, Neptune - These are much larger planets made mostly of gas and liquids.

Key Facts

- The Sun is a medium-sized star that is about 4.6 billion years old
- Earth is the only planet known to support life
- Jupiter is the largest planet, with a mass greater than all other planets combined
- Saturn has the most spectacular ring system
- Mars is called the Red Planet due to iron oxide on its surface

Study Questions

1. What is at the center of our Solar System?

Answer: The Sun is at the center of our Solar System.

2. How many planets are in the Solar System?

Answer: There are eight planets in the Solar System.

3. Which planet is closest to the Sun?

Answer: Mercury is the closest planet to the Sun.

4. Which planet is known as the Red Planet?

Answer: Mars is known as the Red Planet due to iron oxide (rust) on its surface.

5. What are the two main types of planets?

Answer: Terrestrial (rocky) planets and Gas Giant planets.

6. Which is the largest planet?

Answer: Jupiter is the largest planet in our Solar System.

7. Which planet has the most visible rings?

Answer: Saturn has the most spectacular and visible ring system.

8. Which planet can support life?

Answer: Earth is the only planet known to support life.